



Syllabus
Third Year
B.Tech in Electronics and Telecommunication
Engineering
(Programme commenced from AY 2023-24)
(Department of Electronics and Telecommunication
Engineering)

From
Academic Year 2025-26
(SVU-KJSCE 2.0)

(Approved by BOS dated 17th April 2025, FOET dated 22nd
April 2025 and AC dated 28th April 2025, Item No. 14.04)

Preamble

From the Desk of Dean Faculty of Engineering and Technology:

In the era of technological revolution, engineering education must evolve to keep pace with the dynamic demands of industry and society. Our engineering institute is committed to fostering a learning environment that nurtures innovation, creativity, and a profound understanding of engineering principles. The **National Educational Policy 2020 (NEP 2020)** framed by the Government of India recommends a holistic, inclusive, and flexible approach to ensure equitable access to quality education across all levels, promote multidisciplinary research, and impart skill-based education with integration of technology.

Somaiya Vidyavihar, with its esteemed legacy in education, has consistently upheld the values of excellence, inclusivity, and innovation. Applicable for **Somaiya Vidyavihar University (SVU)**'s undergraduate engineering programs, the **SVU Scheme 2023** presented here is aligned with the transformative vision of Somaiya Vidyavihar as well as NEP 2020 to cultivate a holistic, experiential, and interdisciplinary approach to engineering education. The **salient features** of the scheme include:

Professional Core and Elective Courses: The curriculum includes state-of-the-art courses that cover both the fundamentals and emerging trends in respective branches of engineering. With an optimal balance between theoretical knowledge and practical application, core courses provide a strong foundation in essential engineering principles, while elective courses offer flexibility for students to explore and specialize in areas of interest.

Open Elective Courses: Recognizing the importance of interdisciplinary knowledge, the curriculum includes a diverse range of Open Electives categorized into four types: Open Elective Technology (OET), Open Elective Humanities, Open Elective Management (OEHM), and Open Elective Generic (OEG). These courses, offered at institute-level, enable students to expand their knowledge across various disciplines, fostering a versatile skill set and adaptability in an ever-evolving global landscape.

Innovation and Project-based Learning (PBL): The curriculum engages students in innovation and PBL through ideation, mini and major projects right from the first year to the final year of engineering. With diverse projects, collaboration, and field work/community engagement initiatives, students gain a profound understanding of engineering concepts and contribute through innovative solutions to the Sustainable Development Goals (SDGs), societal challenges and advancements.

Learning-by-Doing: The curriculum places emphasis on exposure courses through Skill-Based Learning (SBL) and Activity-Based Learning (ABL), focusing on responsibilities towards society, problem-solving abilities, leadership and teamwork, motivation for life-long learning, etc.

Elements of the Indian Knowledge System: The curriculum incorporates aspects of the Indian Knowledge System that emphasize on drawing insights from ancient wisdom and rich intellectual heritage of India to address modern challenges.

Internships and Research: Enabling students to gain industry insights and enhance their employability, the curriculum integrates flexible internship opportunities in Semester VII or VIII, allowing students to gain hands-on experience in industries, government sectors, NGOs, and MSMEs. Alternatively, they can opt for a specialized research project and courses in Semester VIII. Besides this Semester-long Internship, all the students are required to complete a mandatory 10-week internship over four years, with a maximum of 4 weeks dedicated to socially relevant internships and a minimum of 6 weeks in technical domains.

Learning through MOOCs: The curriculum leverages and promotes Massive Open Online Courses (MOOCs) to offer students flexible and diverse learning opportunities. Complementing on-campus education, students can learn through MOOCs for Open Electives – OET and OEHM during the Pre-final and Final Year, as well as Professional Core courses during their Internship.

Student Exchange Programs: The curriculum also offers student exchange programs that promote global exposure and cross-cultural learning, elevating academic and personal growth. Interested students can participate in the Student Exchange Programs as an alternative to the semester-long internship. Credits from the foreign university where they study will be transferred, providing them with an opportunity to experience different educational systems, cultures, and perspectives.

Minors Courses: Students can expand their academic horizons by pursuing minors in disciplines other than their major, earning additional credits. These minor courses provide an opportunity to acquire multidisciplinary knowledge, significantly enhancing their versatility and adaptability in the professional world.

Honors Courses: For high-achieving students, the SVU 2023 scheme offers Honors courses that delve deeper into specialized topics and gain additional credits for the same. These advanced courses align with high-end industry standards and provide an enriched learning experience, offering multiple opportunities to expand knowledge and expertise in areas of interest.

This forward-thinking SVU 2023 scheme is designed to equip our graduating engineers to emerge as innovative leaders, capable of addressing global challenges and contributing to the advancement of society. Our Boards of Studies, comprising experts in different disciplines, have meticulously designed syllabus for various programs under this SVU 2023 Scheme. We are confident that the joint efforts of the faculty, alumni, students, industry experts, and all the

stakeholders will strengthen the academic, research, and entrepreneurial culture of our institution, reinforcing K. J. Somaiya College of Engineering's position as one of the premier engineering institutions in the nation and a top choice for engineering aspirants

Dr. S. K. Ukarande

Dean – Faculty of Engineering and Technology
Somaiya Vidyavihar University, Mumbai

Preface

From the Desk of Chairman, Board of Studies of Electronics and Telecommunication Engineering

The department of Electronics and Telecommunication (EXTC) was established in 1996. Since its commencement, the primary objective of the department has been to impart quality education, training and promoting research at both undergraduate and postgraduate level. The department focuses on various areas of Electronics and Telecommunication Engineering with broad emphasis on design aspects of electronic and communication systems. The National Board of Accreditation (NBA) has accredited the department thrice from 2005-2008, 2009-2012 and 2013-2015. The college also received the National Assessment and Accreditation Council (NAAC) accreditation in May 2017 and November 2024 with grade “A” in tier I.

EXTC department has all the laboratories with modern equipment. The department has also taken initiative to establish specialized laboratories with industry support in multidisciplinary domains such as networking, RF and Internet of Things. Currently the department has Cisco Centre of Excellence in networking and Texas Innovation Centre under which, various academic & research activities are conducted. The department has always been on a high growth path and has experienced and dedicated faculty members with a strong commitment to engineering education. The major areas of faculty expertise include Basic Electronics, Communication Systems, Computer Networks, Control Systems, Digital Signal Processing, Image Processing, Computer Vision, Microprocessors, Embedded Systems, Instrumentation, RF & Microwaves and VLSI Systems.

More than 450 students are seeking professional education in the department under the supervision and guidance of all faculty and staff members. The UGC conferred autonomy status to college in 2013 and subsequently the college introduced its first autonomous syllabus, KJSCE 2014 from academic year 2014. Furthermore, in the pursuit of imparting quality and holistic education, K. J. Somaiya College of Engineering became a constituent college of Somaiya Vidyavihar University (2019). Accordingly, the department enhanced its curriculum so as to match the requirements of the contemporary world.

The revised curriculum (SVU 2023) is designed with an objective of preparing students to meet new industry challenges and make them aware of the recent advances in technological field. This is achieved by emphasizing on:

- ☐ Strong technical training mechanisms
- ☐ Skill enhancement in design, development, analysis and testing of microelectronic circuits and communication systems
- ☐ Latest trends in simulation and their practices
- ☐ Promoting Interdisciplinary / Multidisciplinary activities through courses and project based learning
- ☐ Contemporary human resource management and development

The curriculum SVU 2023 is apt for students who wish to become professional engineers in the field of Electronics and Telecommunication Engineering thereby finding solutions to practical problems in real world. All theory courses are supported by tutorials / laboratory and special laboratory courses for developing programming skills. Special attention has been given on coverage of fundamental professional core courses from all thrust areas of department. The same are equally supported by a strong laboratory component and considerable project work.

The third year (TY) syllabus in particular covers the basic courses as required in electronics and communication field. It also creates a foundation for interdisciplinary activities as required in today's world. Moreover special attention has been given to utilization of workshop and laboratories by students for the extensive project work which is an additional feature of the program. Thus, with

proper syllabus formulation, we intend to achieve the program objectives, ultimately leading to the attainment of the mission and vision of the department.

HOD EXTC

Board of Studies in Electronics and Telecommunication Engineering

Dr. Bharati. A. Singh	: Chairman
Dr. Kumar Appaiah	: Academician Member
Mr. Shashank Karmarkar	: Industry Member
Dr. Padmaja Joshi	: Research Institute Member
Mr. Prateek Mehta	: Alumni Industry Member
Dr. Ramesh Karandikar	: Faculty Member
Dr. Maruti Zalte	: Faculty Member
Dr. Kiran Ajetroa	: Faculty Member



Our Vision



**Padma Bhushan
Shri Karamshi Jethabhai Somaiya**
Our Founder

Padma Bhushan Shri Karamshi Jethabhai Somaiya's first initiative in education was the founding of a school in 1942 in rural Maharashtra, to provide quality holistic education. It was founded on the belief that education is an important pillar of nation-building with the power to change lives and that it is the duty of the privileged to provide education to the needy.

He later founded the Girivamsi Pragati Mandal, The K J Somaiya Medical Trust, the Girivamsi Education Trust and sister institutions to make great citizens of India and the world. In the words of Swami Vivekananda, "We want that education by which character is formed, strength of mind is increased, the intellect expanded, and by which one can stand on one's own feet." We have now grown into a multi-disciplinary and multi-campus education institution with over 1500 faculty, and 20000+ students.

ज्ञानादेव तु कैवल्यम् ।

KNOWLEDGE ALONE LIBERATES

Our motto is: ज्ञानेन तु कैवल्यम्। Knowledge alone liberates. Liberates from poverty, from hunger. Also to liberate one from the attachments that bind us to small-mindedness. Knowledge also provides opportunity. To make the life lived more meaningful. In the service of one's family, one's community, one's own country, and indeed the world. Bearing in mind that there is no religion other than the life lived in the service of humanity.

न मानुषात् परो धर्मः ।

We will strive to provide access and opportunity to build a more inclusive society.

Our education in any subject will reflect its timeless fundamentals, its current context, and its applications. There is so much scientific discovery taking place, at the intersection of fields, in biology, computing, medicine, the social sciences and everywhere else. We will provide students and faculty with an environment to engage with this world, discover new truths, and make new applications to create and share knowledge.

Our education will also be experiential. With projects that are 'real' and those that complement the learning inside the classroom. Our students and faculty will be at the cutting edge of change, to incubate companies, to create NGOs, and pursue any field of their passion.

Our education will also be holistic. Sports and physical exercise must be a firm part of the curriculum. For students to develop a love for sports, for recreation, for health, for teamwork, and for competition. Our education will also instill an appreciation for art, culture, nature and biodiversity.

अभ्यासेन तु कौन्तेय वैराग्येण च गृह्यते ।

In the Bhagwad Gita, Arjun asks Krishna how is one to control one's mind, which is as fleeting as the wind. Krishna responds that it can only be done through practice and discipline. अभ्यासेन तु कौन्तेय वैराग्येण च गृह्यते । We will strive to teach our students to learn to stay calm in our turbulent world. And our education will also include the ancient Indian tradition, its culture, its depth, and its knowledge. We must keep the connection with our mother tongue and our languages. Languages are storehouses of culture, and the loss of a language takes with it much learning, stored through it over the ages.

Finally, our education will help students lead a full life, and to fall in love with life. Our dream is to build a world-class research and teaching institution, that is global in the reach of its ideas, and universal in its service. Welcome to our community.

Our Mission

To nurture excellence and provide freedom of possibilities in education and research to foster a culture of creativity, innovation, leadership, responsible citizenship, service, and all-round growth.

Vision of the Department:

To become centre of excellence for creating competent engineers with evolving technical skills, leadership qualities with human values to pursue excellence in professional field

Mission of the Department:

Providing quality education to

- Develop technical skills, soft skills and professional ethics to cater the needs of industry.
- Promote research and creativity in Engineering and Technology.
- Inculcate awareness towards societal needs and environmental issues.

Program Educational Objectives (PEO) of the Department:

A graduate of Electronics and Telecommunication Engineering will

PEO1. Excel in professional career by adapting emerging technologies.

PEO2. Pursue higher education, research, address environmental issues with ethical practices.

PEO3. Solve real life problems in a team as a member or as a leader.

Program Outcomes (PO) – Common to all disciplines

- PO1. Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2. Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO3. Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- PO4. Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- PO5. Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- PO6. The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, cultural, environmental, health, safety and legal issues relevant to the professional engineering practice; understanding the need of sustainable development
- PO7. Multidisciplinary Competence:** Recognize/ study/ analyze/ provide solutions to real-life problems of multidisciplinary nature from diverse fields
- PO8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO9. Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- PO10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PO11. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- PO12. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSO)

- PSO1:** Pursue higher studies in the field of Communication, Embedded systems and allied domains
- PSO2:** Acquire, demonstrate industry relevant competencies from core telecommunication discipline and emerging fields

Acronyms used:

1. Acronyms for category of courses and syllabus template

Acronym	Description	Acronym	Description
BS	Basic Science Courses	CA	Continuous Assessment (Theory Course)
ES	Engineering Science	ESE	End Semester Exam
HS	Humanities, Social Sciences and Management Courses	ISE	In- Semester Examination
PC	Professional Core Courses	IA	Internal Assessment
PE	Professional Elective courses	LAB/TUT CA	Continuous Assessment of Laboratory/Tutorial
OET	Open Elective – Technical	TH	Theory
OEHM	Open Elective – Humanities and Management	TUT	Tutorial
OEG	Open Elective Generic	CO	Course Outcome
LC	Laboratory Courses	PO	Program Outcome
PR	Project	PSO	Program specific Outcome
EX	Exposure Course	IKS	Indian Knowledge System

2. Type of Course

Acronym	Description
C	Core Course
E	Elective Course
O	Open Elective Technical
H	Open Elective - Humanities/ Management/
R	Open Elective Generic
P	Project
L	Laboratory Course
T	Tutorial
X	Exposure course
W	Workshop
I	Indian Knowledge System

3. Eight Digit Course code e.g. 216U03C301

Acronym Serially as per code	Description
2	SVU-2023 Second Revision
16	College code
U	Alphabet code for type of program
03/06	Program code/ Common to all
C	Type of course
5	Semester number (Semester V)
01	Course serial number

SEMESTER V (With Effect from 2025-2026)

Teaching and Credit Scheme

Course Code	Course Category	Name of the Course	Teaching Scheme TH-PR-TUT	Total (hrs.)	Credit Scheme TH-PR-TUT	Total Credits
216U03C501	PC	Digital Communication	3 – 0 – 0	3	3 – 0 – 0	3
216U03C502	PC	Computer Communication Networks	3 – 0 – 0	3	3 – 0 – 0	3
216U03C503	PC	Digital Signal Processing	3 – 0 – 0	3	3 – 0 – 0	3
216U03E51X	PE	Departmental Elective - I	3 – 0 – 0	3	3 – 0 – 0	3
216U06Y55X	OE	# Open Elective Humanities Management – 1 (SVU offered / NPTEL or Coursera)	3 – 0 – 0	3	3 – 0 – 0	3
216U03L501	PC	Digital Communication Laboratory	0 – 2 – 0	2	0 – 1 – 0	1
216U03L502	PC	Computer Communication Networks Laboratory	0 – 2 – 0	2	0 – 1 – 0	1
216U03L503	PC	Digital Signal Processing Laboratory	0 – 2 – 0	2	0 – 1 – 0	1
216U03L504	PC	Operating Systems Laboratory	0 – 2 – 0	2	0 – 1 – 0	1
216U03L51X	PE	Elective-I Laboratory	0 – 2 – 0	2	0 – 1 – 0	1
216U03P501	PR	Mini Project	0 – 2 – 0	2	0 – 1 – 0	1
		Total	15 – 12 – 0	27	15 – 6 – 0	21

Students will undertake either one course from SVU baskets offered by other SVU Schools OR Online Course (COURSERA/SWAYAM NPTEL) equivalent to 3 credits

Evaluation Scheme

Course Code	Course Category	Name of the Course	LAB/ TUT CA	CA		ESE	Total
				IA	ISE		
216U03C501	PC	Digital Communication	--	20	30	50	100
216U03C502	PC	Computer Communication Networks	--	20	30	50	100
216U03C503	PC	Digital Signal Processing	--	20	30	50	100
216U03E51X	PE	Departmental Elective - I	--	20	30	50	100
216U06Y55X	OE	Open Elective Humanities Management - 1		20	30	50	@100
216U03L501	PC	Digital Communication Laboratory	50	--	--	--	50
216U03L502	PC	Computer Communication Networks Laboratory	50	--	--	--	50
216U03L503	PC	Digital Signal Processing Laboratory	50	--	--	--	50
216U03L504	PC	Operating Systems Laboratory	50	--	--	--	50
216U03L51X	PE	Elective-I Laboratory	50	--	--	--	50
216U03P501	PR	Mini Project	50				50
		Total	300	100	150	250	800

@ Students will undertake the examination/evaluation as per the opted course from SVU basket OR Online (COURSERA/ SWAYAM NPTEL) Course. The credits will be considered for credit transfer after successful completion of the chosen course(s). For basket course the evaluation will be scaled to 100 marks. The grades/marks obtained in online courses will not be considered for CGPA calculation

Departmental Elective -I			
Course Code	Name of Elective	Course Code	Name of Elective
216U03E511	Random Processes	216U03E512	Controls systems and Applications
216U03E513	Database Management System	216U03E514	Soft Computing
216U03E515	Machine Learning		

SEMESTER VI (With Effect from 2025-2026)

Teaching and Credit Scheme

Course Code	Course Category	Name of the Course	Teaching Scheme TH-PR-TUT	Total (hrs.)	Credit Scheme TH-PR-TUT	Total Credits
216U03C601	PC	Wireless Communication	3 – 0 – 0	03	3 – 0 – 0	03
216U03C602	PC	RF and Antennas	3 – 0 – 0	03	3 – 0 – 0	03
216U03C603	PC	Optical fibre Communication	3 – 0 – 0	03	3 – 0 – 0	03
216U03E61X	PE	Departmental Elective-II	3 – 0 – 0	03	3 – 0 – 0	03
216U06Y55X	HS	# Open Elective Humanities Management - II	3 – 0 – 0	03	3 – 0 – 0	03
216U03L601	PC	Wireless Communication Laboratory	0 – 2 – 0	02	0 – 1 – 0	01
216U03L602	PC	RF and Antennas Laboratory	0 – 2 – 0	02	0 – 1 – 0	01
216U03L603	PC	Optical Fibre Communication Laboratory	0 – 2 – 0	02	0 – 1 – 0	01
216U03L61X	PE	Elective- II Laboratory	0 – 2 – 0	02	0 – 1 – 0	01
216U03P601	PR	Project- I	0 – 4 – 0	04	0 – 2 – 0	02
		Total	15 – 12 – 0	27	15 – 06 – 0	21

Students will undertake either one course from SVU baskets offered by other SVU Schools OR Online Course (COURSERA/SWAYAM NPTEL) equivalent to 3 credits.

Evaluation Scheme

Course Code	Course Category	Name of the Course	LAB/ TUT CA	CA		ESE	Total
				IA	ISE		
216U03C601	PC	Wireless Communication	--	20	30	50	100
216U03C602	PC	RF and Antennas	--	20	30	50	100
216U03C603	PC	Optical fibre Communication	--	20	30	50	100
216U03E60X	PE	Departmental Elective-II	--	20	30	50	100
216U06Y55X	HS	Open Elective Humanities Management - II		20	30	50	@ 100
216U03L601	PC	Wireless Communication Laboratory	50				50
216U03L602	PC	RF and Antennas Laboratory	50	--	--	--	50
216U03L603	PC	Optical Fibre Communication Laboratory	50	--	--	--	50
216U03L60X	PE	Elective- II Laboratory	50	--	--	--	50
216U03P601	PR	Project- I	50	--	--	--	50
			250	100	150	250	750

@ Students will undertake the examination/evaluation as per the opted course from SVU basket OR Online (COURSERA/ SWAYAM NPTEL) Course. The credits will be considered for credit transfer after successful completion of the chosen course(s). For basket course the evaluation will be scaled to 100 marks. The grades/marks obtained in online courses will not be considered for CGPA calculation

Departmental Elective - II			
Course Code	Name of Electives	Course Code	Name of Electives
216U03E611	Basic VLSI Design	216U03E612	Instrumentation system
216U03E613	Image Processing	216U03E614	Cyber Security
216U03E615	Cloud Computing		

Note:

As per college internship policy, It is mandatory for every student to complete 10 weeks of Internship spanning over the four years of B.Tech Programme over and above the academic credits. Students can take up internships in community services / socially relevant projects (optional and limited to 4weeks) and in the technical domain (minimum 6 weeks or more). Students will be awarded an internship completion certificate along with their graduation.

Course-wise Detailed Syllabus

SEMESTER V

Course Code	Name of the Course			
216U03C501	Digital Communication			
Teaching Scheme	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03
Evaluation Scheme	Marks			
	LAB/TUT CA	CA (TH)		ESE
		IA	ISE	
	--	20	30	50
				100

Course pre-requisites:

- Communication Systems
- Basics of probability theory

Course Objectives:

The objective of the course is to impart knowledge of digital communication systems with emphasis on digital modulation techniques. It helps in understanding the basic building blocks of a digital communication system and analyse the signal flow in the system. The course analyses error performance of a digital communication system. It introduces the concept and basics of information theory and the basics of source and channel coding /decoding.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Apply information theory in source coding
- CO2.** Analyse distortion less baseband transmission and reception of signal along with waveform coding techniques
- CO3.** Analyse different band pass digital transmission and reception of signal.
- CO4.** Apply concept of sampling for reliable communication.
- CO5.** Apply different methods of error control coding and schemes required for reliable transmission of digital communication.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs	CO
1	Sampling Process and types of sampling		02	CO1
	1.1	Sampling theorem for low pass & band pass signal, Ideal sampling process, Spectrum of sampled signal, Reconstruction of sampled signal and amplitude response of reconstruction filter.(No derivation)		
	1.2	Signal distortion in sampling, signal-to-distortion Ratio, practical aspect of sampling, flat top sampling, spectrum of flat top sampled signal.		
2	Information Theory , Source coding and Channel capacity		08	CO2
	2.1	Basic Signal Processing Operations in Digital Communications, Uncertainty, Information, and Entropy		
	2.2	Source Coding Theorem, Shannon Fano source coding, Huffman source coding , Lempel –Ziv		
	2.3	Channel capacity, Channel coding theorem, Channel Capacity Theorem		
3	Baseband Modulation and Demodulation Techniques		08	CO3
	3.1	Signal representation: GRAM-SCHMIDT Orthogonalization procedure, Geometric representation of signals		
	3.2	Baseband data formats and their power spectral density: NRZ unipolar, NRZ polar, NRZ bipolar, Manchester, M-ary PAM signals (derivations not expected)		
	3.3	Baseband signal detection and estimation: Maximum likelihood detection and calculation of its probability of error, matched filter receiver , correlation receiver		
	3.4	Nyquist criteria for distortion less baseband transmission: Intersymbol interference(ISI),correlative coding- Duobinary signaling, eye pattern,equalization for efficient baseband data transmission		
		#Self learning topic: non-coherent digital communication systems		
4	Bandpass Modulation and Demodulation		12	CO4
	4.1	Bandpass digital transmitter and receiver model, digital modulation schemes		
	4.2	Generation, detection, spectrum, bandwidth efficiency, signal space diagram and probability of error analysis of Phase Shift Keying: BPSK, QPSK, DPSK, DEPSK Amplitude Shift Keying: BASK Frequency Shift Keying: BFSK, Quadrature Amplitude Modulation (QAM), Minimum Shift Keying (MSK) , Gaussian MSK		

5	Error Control Coding		
	5.1	Linear block codes, Hamming code, generator matrix, systematic linear block codes, parity check matrix, syndrome testing, error correction and decoder implementation	15
	5.2	Cyclic codes: Algebraic structure of cyclic codes, binary cyclic code properties, encoding in systematic form, circuits for dividing polynomials, systematic encoding with shift register and error detection.	
	5.3	Convolutional coding and decoding, State and Trellis Diagram, Viterbi algorithm Overview of Turbo code, Low Density Parity Check codes	
		Total	45

Students should prepare self-learning topics on their own. Self – learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in IA and or laboratory experiments.

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Haykin Simon	<i>Digital Communication Systems</i>	John Wiley and Sons, India	1 st edition 2013
2	Taub Schilling	<i>Principles of Communication systems</i>	Tata McGraw Hill, India	4 th Edition 2013
3	T. L. Singal	<i>Digital Communication</i>	McGraw Hill Education , India	1 st Edition 2015
4	Bernard Sklar and Pabitra Kumar Ray	<i>Digital communications</i>	Pearson Education	2 nd Edition 2014
5	Ranjan Bose	<i>Information theory coding and cryptography</i>	McGraw Hill Education , India	3 rd Edition 2016

Course Code	Name of the Course				
216U03C502 ^{\$}	Computer Communication Networks				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

216U03C502^{\$} - The course is common with 216U41C502 of CCE program.

Course pre-requisites:

- Communication Systems
- Basics of Digital communication

Course Objectives: The objective of this course is to enable the students to understand the fundamentals of Computer Networks, OSI layers and TCP/IP model, networking devices. It introduces the concept of IP addressing and subnetting and Routing . The course covers the protocols in various TCP/IP layers such as MAC protocols, ARP, Internet Protocol ,TCP,UDP, DNS, HTTP and Email protocols etc.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Understand layered architecture of communication network, networking devices and the physical layer.
- CO2.** Explore the significance of data link layer ,error and flow control and MAC protocols
- CO3.** Understand and apply IP addressing, routing algorithms, and design complete network and its related security as per the given requirements.
- CO4.** Understand functionalities of transport layer, concept of port numbers and UDP and TCP protocols with their operations
- CO5.** Understand and demonstrate various application layer protocols as HTTP,DNS,SMTP

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs	CO
1	Introduction to Data Communication Networks and Physical Layer		06	CO1
	1.1	Network Criteria, Types of Networks: LAN, WAN, MAN. Concept of Packet switching and Circuit switching.		
	1.2	Need for layered architecture, OSI model and TCP/IP reference model		
	1.3	Networking Devices: Hub, Bridge, Router, Switch		
	1.4	Transmission media: Guided and Unguided media		
2	The Data Link Layer		11	CO2
	2.1	Introduction to Data link layer, Frame structure (bit,byte and character stuffing) Error control and Flow control. Error detection : Parity check, Hamming Code CRC and Checksum		
	2.2	Flow control Protocols: Stop-and-wait, Go-Back-N, Selective-Repeat, Piggybacking		
	2.3	Multiple Access Protocols: Controlled Access, Channelization, Random Access protocols: ALOHA, slotted ALOHA, CSMA, CSMA/CD, CSMA/CA.		
	2.4	Concept of Virtual LAN (VLAN), Overview of Ethernet Standards		
3	The Network Layer		17	CO3
	3.1	Network layer Services, IPv4 addressing , Subnetting		
	3.2	Internet Protocol (IPv4)-Header format ,Fragmentation in IP		
	3.3	Private and Public IP addresses, Network Address Translation (NAT)		
	3.4	Routing : Distance vector routing, Link State routing. Unicast routing protocols: Overview of RIP , OSPF, BGP		
	3.5	Address Resolution Protocol(ARP), RARP, Dynamic Host Configuration protocol(DHCP), ICMP , IPv6 addressing		
	3.6	Introduction to Network Security: Security Goals, Attacks Services, Techniques Network layer security : IPSec protocol, Two Modes: Tunnel and Transport ,Services Provided by IPSec		
4	The Transport Layer		05	CO4
	4.1	Transport Layer services, Connectionless and connection oriented services, Process-to Process connection, concept of port numbers and socket address.		
	4.2	User Datagram Protocol (UDP) ,Transmission Control Protocol (TCP), TCP connection using 3 way Handshaking , TCP Timers		
	4.3	Introduction to SCTP		

5	Application Layer		06	CO5
	5.1	Application Layer services, HTTP- Hyper Text Transfer Protocol, HTTPs		
	5.2	File Transfer Protocol –FTP, Telnet, SCP		
	5.3	Domain Name System-DNS		
	5.4	Email Protocols: SMTP ,PoP3 ,IMAP4		
Total			45	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	B. Forouzan,	<i>Data Communication and Networking</i>	McGraw Hill Publication, India	5th Edition, 2013
2	L. Garcia et al	<i>Communication Networks</i>	McGraw Hill Publication, India	2nd Edition, 2004
3	J. F. Kurose and K. W. Ross	<i>Computer Networking: A Top-Down Approach</i>	Pearson Publication, India	5th Edition, 2012

Course Code	Name of the Course				
216U03C503	Digital Signal Processing				
Teaching Scheme	TH	P	TUT	Total	
	--	--	--	--	
Credits Assigned	03		--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
		20	30	50	100

Course pre-requisites:

- Signals and systems
- Mathematics for Communication Engineering

Course Objectives:

Signal processing is an area of science and engineering developed over decades. This fundamental course addresses the representation, analysis, and design of discrete time Systems. The major concepts covered include Discrete-time processing of continuous-time signals in time and frequency domain, discrete Fourier transform, FFT algorithm, structures for DT systems, design techniques for recursive (IIR) and non-recursive (FIR) filters; multirate techniques.

Course Outcomes (CO):

Course Outcomes

At the end of successful completion of the course the student will be able to

- CO1.** Analyze discrete time systems in time and Z-domain
- CO2.** Realize discrete system using different structures
- CO3.** Analyze discrete time systems using discrete Fourier transform
- CO4.** Design Finite Impulse Response and Infinite Impulse Response filters
- CO5.** Understand concept of Multi-rate signal processing

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs	CO
1	Introduction: Discrete Time (DT) systems and analysis in time and Z-domain		10	CO1 CO2
	1.1	Review of discrete time signals and systems: Concept of auto-correlation, cross-correlation, Linear and circular convolution, Impulse response, transfer function.		
	1.2	LTI system as Frequency selective filters: Ideal filter characteristics, Low pass, High pass, Band pass, band stop, Notch, resonator, averaging, comb, all pass filters. Pole Zero placement and analysis, Structures for realization of systems: Direct form I and II, cascade, parallel.		
	1.3	Introduction to DSP systems and their applications in areas such as Biomedical and Communication systems		
2	Analysis of discrete time system in frequency domain		10	CO3
	2.1	Discrete Time Fourier Transform (DTFT) : analysis of Linear Time Invariant (LTI) DT systems using DTFT.		
	2.2	Discrete Fourier Transform (DFT): Frequency domain sampling, Properties of DFT, Filtering of long Data sequences, Analysis of LTI DT system using DFT, aliasing in frequency domain.		
	2.3	Fast Fourier Transform (FFT): Efficient computation of DFT, Radix 2 Decimation In Time (DIT) and Decimation In Frequency (DIF) FFT algorithms for 8-point DFT computation.		
3	Design of FIR Filters		09	CO4
	3.1	Characteristics of FIR filters with linear phase, symmetric and antisymmetric sequences.		
	3.2	Design specification of digital filters.- FIR filter design using windowing techniques : Design with rectangular, Hamming, and Hanning window		
	3.3	FIR filter design using frequency sampling method.		
4	Design of IIR Filters		10	CO4
	4.1	Frequency response of analog and digital IIR filter, Analog Butterworth and Chebyshev-1 filters (No derivation for transfer function, Frequency transformations- Analog, digital		
	4.2	Low pass digital IIR filter design using Impulse Invariance and Bilinear Transformation method using Butterworth and Chebyshev-1 approximation		

5	Multi rate Digital Signal Processing		06	CO5
	5.1	Decimation and Interpolation by integer factor		
	5.2	Sampling rate conversion, Poly phase decomposition		
Total			45	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1.	Sanjit Mitra	<i>Digital Signal Processing- A computer Based Approach</i>	TATA McGraw Hill, India	4th Edition, 2013
2.	A.V. Oppenheim	<i>Digital Signal Processing</i>	Prentice Hall, India	2nd Edition, 2013
3.	John Proakis, Dimitris Manolakis	<i>Digital Signal Processing, Principles, Algorithms and Applications</i>	Prentice Hall, India	4th Edition, 2005
4.	A. Nagoor Kani	Digital Signal Processing	TATA McGraw Hill, India	2nd Edition, 2013
5.	Ashok Ambardar	<i>Digital Signal Processing: A Modern Introduction</i>	CENGAGE Learning India	1st Edition, 2004

DEPARTMENTAL ELECTIVE I

Course Code	Name of the Course				
216U03E511	Random Processes				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites:

- Concepts of probability
- Signals and systems

Course Objectives:

The objective of this course is to give a clear understanding of random processes on the basis of statistics of random variables. Course contents include definition and statistics of discrete time and continuous time random processes specific random processes and all types of stationary random processes. The depth is extended to analysis and applications of random processes. Thus by doing this course, students should be able to apply statistics knowledge to solve any practical problem in the areas of engineering and also the day to day life.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Understand the statistics of single and multiple random variables.
- CO2.** Understand the definition and statistics of random processes.
- CO3.** Understand the definition and statistics of specific random processes.
- CO4.** Analyse the random processes.
- CO5.** Apply concepts of statistics and random process for problem solving in engineering and day to day life

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs	CO
1	Review of probability		02	CO1
	1.1	Definitions - Sets, Fields, Sample space and Events, Probability, Axioms, Joint and Conditional probabilities, Independence, Total probability		
2	Random variables		14	CO1 CO5
	2.1	Definitions-Random variables, Continuous and discrete random variables, Cumulative distribution functions of continuous and discrete random variables, Probability mass function (pmf), Probability density functions(pdf) and properties, Expectation, Variance and moments of a random variable		
	2.2	Function of a random variable : Pdf of the function of a random variable, expected value of function of a random variable, variance of function of a random variable		
	2.3	Multiple random variables: One function of two random variables, two functions of two random variables, joint density and distribution functions, conditional density and distribution functions, conditional expectation, independence of random variables, conditional moments, uncorrelated random variables, independence of random variables, covariance and correlation		
3	Random Process		14	CO2 CO3
	3.1	Introduction to random processes: Definition of a random process, discrete time and continuous time random processes Probabilistic description of random processes by first order and second order statistics- mean, variance, covariance and autocorrelation, joint distribution Specific discrete time and continuous time random Processes Bernoulli process, Binomial process, Poisson process, Gaussian process Multiple random Processes-Relationship between two random processes, cross-correlation, cross-covariance		
	3.2	Stationary random processes: Condition for stationarity of a random process, wide sense stationary process (WSS) , strict sense stationary process (SSS), Examples of WSS and		
	3.3	Ergodicity of random process: Ensemble average and time average of a random process, mean- ergodic random process, correlation-ergodic random process		
4	Analysis of random process		08	CO4
	4.1	Power spectral density: Power spectral density of continuous time random processes, power spectral density of discrete time random processes, power spectral density as a time average		

	4.2	Examples of random processes: White noise process and white noise sequence, Gaussian process, Poisson process, Markov Process		
5	Markov Chains			
	5.1	Discrete- time Markov chains: Transition probability matrix, nstep transition probabilities, Chapman-Kolmogorov equations, Steady state probabilities	07	CO5
	5.2	Continuous-time Markov chain: State probabilities, Steady state probabilities, Global balance equations, Birth-and-death process		
Total			45	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Alberto Leon-Garcia	<i>Probability and Random Processes for Electrical Engineering</i>	Pearson, UKcountry of origin	2 nd Edition, 1993
2	Geoffrey Grimmett, David Stirzaker	<i>Probability and random processes</i>	Oxford University Press inc., published in United States	3 rd Edition, 2001
3	H. Stark and J.W Woods	<i>Probability and Random Processes with Applications to Signal Processing</i>	Pearson, UKcountry of origin	4 th Edition, 2011
4	Hwei P. Hsu	<i>Schaum's outline of Probability Random Variables and Random Process</i>	Mcgraw-Hill inc., USA	4 th Edition, 2019

Course Code	Name of the Course			
216U03E512	Control Systems and Applications			
Teaching Scheme	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03
Evaluation Scheme	Marks			
	LAB/TUT CA	CA (TH)		ESE
		IA	ISE	
	--	20	30	50
				100

Course pre-requisites:

- Dynamics, differential equations and Laplace transforms

Course Objectives:

Objectives of this course are: (paragraph)

- To understand the fundamental concepts of control systems and mathematical modeling of the system.
- To explore the concept of time response and frequency response of the system.
- To learn basics of stability analysis of the system.
- (To do)case study for any feedback system through activity based learning.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Derive the mathematical model of different types of the systems.

CO2. Understand the basics of state variable models of system.

CO3. Explore analysis of the systems in time domain.

CO4. Explore analysis of the systems in frequency domain.

CO5. Design a feedback system.

Detailed Syllabus

Module No.	Unit No.	Details	Hrs	CO
1	Introduction to Control System Analysis		12	CO1
	1.1	Introduction: Open loop and closed loop systems, feedback and feed forward control structure, control systems models and examples of control systems		
	1.2	Dynamic Response: Standard test signals, transient and steady state behavior of first and second order systems, steady state errors in feedback control systems and their types		
	1.3	Transfer Function Models of Various Systems: Models of mechanical systems, models of electrical systems, block diagram reduction, signal flow graph and Mason's gain rule		
2	State Variable Models of LTI Systems		08	CO2
	2.1	State Variable Models of Various Systems: State variable models of mechanical and electrical systems		
	2.2	State Transition Equation: Concept of state transition matrix, properties of state transition matrix, solution of homogeneous and non-homogeneous systems, concept of eigenvalues and eigenvectors, controllability and observability		
3	Stability Analysis in Time Domain		10	CO3
	3.1	Concepts of Stability: Concept of absolute, relative and robust stability, Routh stability criterion, lag and lead compensator <i>#self-study: PID controllers</i>		
	3.2	Root Locus Analysis: Root-locus concepts, general rules for constructing root-locus, Root-locus analysis of control systems		
4	Stability Analysis in Frequency Domain		10	CO4
	4.1	Introduction: Frequency domain specifications, response peak, peak resonating frequency, relationship between time and frequency domain specification of system and stability margins.		
	4.2	Bode Plot: Magnitude and phase plot, method of plotting Bode plot, stability margins on the Bode plots		
	4.3	Nyquist Criterion: Polar plots, Nyquist stability criterions, Nyquist plot, gain and phase margins		
5	Applications of Control Systems		05	CO5
	5.1	Concepts and Block Diagrams: Servo, Microcontroller, Robotic, Fuzzy, IoT, Communication systems and AI systems		
	5.2	Introduction to Programmable logic controller (PLC): ladder programming, architecture		
Total			45	

*There will be IA as mini-project of 20 marks

Students should prepare self-learning topics on their own. Self – learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in IA and or laboratory experiments.

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	M. Gopal	<i>Control Systems: Principle and Design</i>	Tata McGraw Hill, India	Fourth Edition, eighth reprint 2015
2.	Benjamin C. Kuo	<i>Automatic Control Systems</i>	Prentice Hall of India	Seventh Edition, 1995
3.	M.N. Bandyopadhyay	<i>Control Engineering.</i>	Prentice Hall of India	Second Edition, 2004
4.	J.Nagrath and M.Gopal	<i>Control System Engineering</i>	New Age International Publishers, India	Firth Edition, 2007

Course Code	Name of the Course				
216U03E513	Database Management System				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Data Structure and Programming Logic

Course Objectives: This course provides a comprehensive understanding of database management systems. Students will learn core concepts, including data models, database architecture, and the role of DBMS in information systems. The course covers SQL for data definition, manipulation, and querying, as well as relational database design using ER models and normalization. Topics include database implementation, management, performance optimization, security, transaction management, concurrency control, and recovery.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Understand the role of DBMS in real-world applications and differentiate it from traditional file systems.
- CO2.** Understand various data models and apply ER modeling to design diagrams for real-world scenarios.
- CO3.** Understand the internal storage structures in a physical DB design
- CO4.** Understand indexing and hashing techniques, analyze query processing steps and cost, and apply query optimization strategies to improve database performance
- CO5.** Apply fundamental concepts of transaction processing, comparison with non-relational database and utility of it.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs	CO
1	Introduction to Databases:		06	CO1
	1.1	Introduction, Purpose of Database Systems, DBMS system architecture , Data Models, Data independence		
	1.2	Database languages, Database Users and Administrator, Different types of Database Systems		
2	Database Models and SQL		09	CO2 CO3
	2.1	Database design phases, E-R Model, Constraints, E-R Diagrams, E-R design issues		
	2.2	Entity set, Extended E-R features		
	2.3	Relational model concepts, Constraints Relational Algebra: Unary, Binary and Set theory relational operations Data definition commands: attribute constraints, SET operations, Aggregate functions, Null Values, Nested sub queries, complex queries, Views Data control commands Data manipulation commands: Insert, Update, Joined relations Integrity and security: Domain constraints, Referential integrity, Triggers; Security and Authorization in SQL		
3	Relational Database Design		10	CO4
	3.1	Design guidelines for relational schemas, Functional Dependencies, Armstrong Axiom		
	3.2	First Normal form, Second Normal form, Third normal Form.		
	3.3	Decomposition and Desirable properties of decomposition, decomposition using multivalued dependencies, fourth normal form.		
	3.4	Overall database design		
4	Indexing, Hashing , Query processing and Optimization		10	CO4
	4.1	Basic concepts, ordered indices: dense and sparse, multilevel indices, secondary indices		
	4.2	Hashing: Static hashing, dynamic hashing, comparison of ordered indexing and hashing		
	4.3	Query processing: Steps involved in query processing, measures of query cost, algorithms for SELECT and PROJECT operations.		
	4.4	Query Optimization: Overview, Transformation of relational expressions, Estimating statistics, Choice of evaluation plan		

5	Transactions and Non-relational databases		10	CO5
	5.1	Transaction Concepts, Transaction state, ACID properties, concurrent executions.		
	5.2	Distributed Databases: Sharding and Replication, Consistency, Master Slave Replication, Peer to Peer replication, Introduction to cache, The CAP Theorem, NoSQL Data Models		
	5.3	NoSQL approach, Key-Value and Document Data Models, Column-Family Stores, Comparison of relational databases to new NoSQL stores, MongoDB, Cassandra		
Total			45	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Korth, Slberchatz, Sudarshan	<i>Database System Concepts</i>	McGraw Hill Education, India	7 th Edition, 2018
2	Raghu Ramakrishnan and Johannes Gehrke	<i>Database Management Systems</i>	McGraw Hill Education, India	3 rd Edition 2002
3	Elmasri and Navathe	Fundamentals of Database Systems	Pearson Education, India	7 th Edition, 2019
4	G.K. Gupta	<i>Database Management Systems</i>	McGraw Hill Education, India	6 th Edition, 2011

Course Code	Name of the Course				
216U03E514	Soft Computing				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites:

- Calculus, Linear Algebra, Probability & Statistics

Course Objectives :

- Understand Soft Computing concepts, technologies, and applications
- Understand the underlying principle of soft computing with its usage in various applications.
- Understand different soft computing tools to solve real life problems.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.**Comprehend the concepts of biological neurons and artificial neurons
CO2.Create different neural networks of various architectures
CO3.Apply the fundamental principles of mathematics for learning neural networks
CO4.Understand the concept of Fuzzy logic, fuzzy rules and fuzzy reasoning
CO5.Solve single-objective optimization problems using GAs.

Detailed Syllabus

Module No.	Unit No.	Details	Hrs.	CO
1	Introduction to Soft computing and Neural Networks		08	CO1 CO2
	1.1	Introduction to Soft Computing, Hard and Soft Computing, Major Areas of Soft Computing, Applications of Soft Computing		
	1.2	Biological neurons and its working, Simulation of biological neurons to problem solving, perceptron, activation functions, Different ANNs architectures.		
	1.3	Concepts of supervised, unsupervised learning, cost functions, hypotheses, Linearly separable and linearly non-separable systems and their examples, review of fundamental learning techniques.		
2	Back propagation learning		10	CO2 CO3
	2.1	Backpropagation algorithms, Gradient Descent (GD), Momentum Based GD, Adagrad, AdaDelta, Adam optimizer		
	2.2	Bias variance tradeoff, L2 regularization, Early stopping, Dataset augmentation, Dropout		
3	Special Network and models		11	CO2 CO3
	3.1	Radial basis functions, Topologically Organized Neural Networks – Competitive Learning, Kohonen Maps		
	3.2	Convolutional Neural Networks (CNN), Architectures, AlexNet, MobileNet, transfer learning.		
	3.3	Neural Networks as Associative Memories - Hopfield Networks, Long Short-Term Memory		
		Self Study: Transformer		
4	Fuzzy logic			
	4.1	Basic Fuzzy logic theory, sets and their properties, Operations on fuzzy sets	08	CO4
	4.2	Fuzzy relation and operations on fuzzy relations and extension principle, Fuzzy membership functions and linguistic variables		
	4.3	Fuzzy rules, Fuzzification and defuzzification and their methods		
	4.4	Fuzzy inference systems, Mamdani Fuzzy models, and Fuzzy knowledge based controllers, Fuzzy Classification.		

5	Genetic Algorithms			
	5.1	Concept of "Genetics" and "Evolution" and its application to probabilistic search techniques, Basic GA framework and different GA architectures.	08	CO5
	5.2	GA operators: Encoding, Crossover, Selection, Mutation, etc. Solving single-objective optimization problems using GAs.		
Total			45	

Reference Books*

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Jacek M. Zurada	<i>Introduction to Artificial Neural Systems</i>	Jaico Publishing House, India	1992
2.	Ian Goodfellow, Yoshua Bengio, Aaron Courville	<i>Deep Learning</i>	MIT Press	3 January 2017
3.	Geron Aurelien	<i>Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems</i>	O'Reilly	4th edition, 2017
4.	Thimothy J. Ross	<i>Fuzzy Logic with Engineering Applications</i>	Wiley, India	Third edition (2011)
5.	David E. Goldberg	<i>Genetic Algorithms in Search, Optimization, and Machine Learning</i>	Pearson Education	1st edition (2008)

Course Code	Name of the Course				
216U03E515	Machine Learning				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites:

- Basics of Data Analysis
- Data Structures & Algorithms

Course Objectives:

Machine learning is a science of learning from data using the fields of Statistics, Linear Algebra, Computer Programming and Optimization, to make prediction or classification. Its use is rising these days with lot of applications in all industries. It is feasible due to higher information processing and memory processing ability of today's computing systems. The course will introduce students with basic models and algorithms for machine learning, and prepare students for research or industry application of machine learning techniques.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Learn process to develop models from data.
- CO2.** Learn supervised and unsupervised machine learning algorithms.
- CO3.** Learn models evaluation.
- CO4.** Understand feature and dimensionality reduction techniques.
- CO5.** Apply the algorithms to solve a real-world problem.

Detailed Syllabus

Module No.	Unit No.	Details	Hrs.	CO
1	Introduction		10	CO1
	1.1	Importance of machine learning, Data diversity-structured, unstructured, cross sectional, Time series; Examples of applications in diverse fields		
	1.2	Forms of learning- Supervised, Unsupervised, Reinforcement; Steps in developing Machine Learning algorithm		
	1.3	Regression versus Classification, Binary and Multi-class classification, Metrics for assessing regression performance- Mean Square Error, Mean Absolute Error; Metrics for assessing classification performance- Misclassification error, Confusion matrix, Receiver Operating Characteristics(ROC), Sensitivity, Specificity		
	1.4	Data preparation: Exploratory data analysis- bar chart, histogram, box plot, scatter plot, heat map, contour map, radar plot; Handling missing values- mean/median, k-nn; Data transformation-one hot encoding, log-transform, normalization		
	1.5	Model selection: Definition of a good model, overfitting and generalization, bias-variance tradeoff, regularization, cross validation techniques- Validation set method, k-fold cross validation, Leave One Out cross validation		
2	Supervised Learning 1		12	CO2 CO3 CO5
	2.1	Linear regression: Simple linear regression, loss function, data distribution, models, parameters, hyperparameters, optimization algorithms, hypothesis testing of regression model, residual analysis, regularization, shrinkage methods- LASSO, RIDGE, issues in regression- feature scaling, categorical features, outliers; Multiple linear regression, collinearity; Non-linear regression; Logistic regression for binary and multiclass classification		
	2.2	k-nearest neighbor(k-nn): Regression and classification with k-nn, distance measures, comparison between LR, LogR, k-nn; Issues in k-nn- curse of dimensionality, computational efficiency		
	2.3	Decision trees: Classification trees, regression trees; C4.5, CART; Regularization and pruning		
	2.4	Bayes' Rule and Naive Bayes classifier		

3	Supervised Learning 2		10	CO2 CO3 CO5
	3.1	Support Vector Machine: Construction of the max- margin classifier; Hard vs. soft margin SVM, slack variables; Hinge loss; Introduction to dual Optimization, SVM Dual; Kernel trick- feature combinations and kernels, kernel SVM; SVM for regression, binary and multiclass classification		
	3.2	Neural Networks: Representation-Perceptron, single and multilayer networks, Neutral networks for regression, binary and multi-class classification; Learning-Back propagation algorithm, Regularization		
4		Unsupervised Learning	06	CO2 CO3 CO5
	4.1	Clustering: Similarity\Distance measures, k- means clustering ,Hierarchical clustering		
	4.2	Gaussian Mixture Models, Expectation-Maximization		
5		Ensembles and Dimensionality Reduction	07	CO4 CO5
	5.1	Boosting and Ensemble methods: Bagging, Boosting, random forest; Combining multiple models		
	5.2	Feature selection: Forward selection and Backward selection; Dimensionality reduction: Principal Component Analysis		
Total			45	

Reference Books*

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Trevor Hastie, Robert Tibshirani and Jerome Friedman	<i>The Elements of Statistical Learning</i>	Springer Series in Statistics	Second edition, 2009
2.	Ethem Alpaydin	<i>Introduction to Machine Learning</i>	PHI	Third edition, 2015
3.	M. Gopal	<i>Applied Machine Learning</i>	PHI	First edition, June 2019
4.	Shai Shalev-Shwartz, Shai Ben-David	<i>Understanding Machine Learning: From Theory to Algorithms</i>	Cambridge University Press	First edition, 2014

Course Code	Name of the Course				
216U03L501	Digital Communication Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

*Distribution of marks:

Instructions:	- LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8 and maximum 10 experiments will be performed in a semester. - LAB CA also includes evaluations covering entire syllabus (25 marks). - LAB CA schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge.
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Course Code	Name of the Course				
216U03L502	Computer Communication Networks Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
		50	--	--	--

Instructions:	• LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8 experiments will be performed in a semester. • LAB CA also includes evaluations covering entire syllabus (25 marks). • LAB CA schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge.
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Course Code	Name of the Course				
216U03L503	Digital Signal Processing Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

*Distribution of marks:

Instructions:	- LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8-10 experiments / case studies will be performed in a semester. - LAB CA also includes Evaluations based on entire syllabus (25 marks). - LAB CA evaluations schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge.
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Course Code	Name of the Course				
216U03L504	Operating Systems Laboratory				
Teaching Scheme	TH	P	TUT	Total	
	--	02		02	
Credits Assigned	--	01		01	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

Course pre-requisites: Nil

Course Objectives: This course gives general description of the basic concepts of operating systems, functionalities and objectives, structure and components. It helps in understanding the services provided by and the design of an operating system. It helps to understand what a process is and how processes are synchronized and scheduled. The course covers the structure and organization of the file system, memory management, I/O management.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO 1.** Understand the role of an operating system, its function and issues.
- CO 2.** Compare different algorithms used for management and scheduling of processes.
- CO 3:** Understand different approaches to memory management.
- CO 4:** Understand the structure and organization of the file system.
- CO 5:** Explore I/O systems, Storage structure and Unix system.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs	CO
1	Fundamental of Operating System(OS)			
	1.1	Definition, objectives, functions, services, types, and different views of OS.	04	CO1
	1.2	Operating System as a resource manager, system calls, and types of shells. Monolithic systems, layered systems, client server model, monolithic kernel and microkernel.		
		#Self-Study: Comparison of different types of operating systems in terms of OS structures.		
2	Process Management			
	2.1	Different states of a Process, Process State transition diagram. Thread: Definition, Various states, Types of threads, Concept of multithreads. Inter-process Communication: Concurrent processes, Critical Section, Race Conditions, Mutual Exclusion, Hardware Solution, Semaphores, Peterson's Solution, The Producer / Consumer Problem.	10	CO2
	2.2	Deadlocks: Definition, Necessary and sufficient conditions for Deadlock, Deadlock Prevention, and Deadlock Avoidance: Banker's algorithm, Deadlock detection and Recovery.		
	2.3	Types of Schedulers, Scheduling criteria: CPU utilization, Throughput, Turnaround Time, Waiting Time, Response Time. Scheduling algorithms: Pre-emptive and non-pre-emptive, FCFS, SJF, Round-robin.		
		#Self-Study: Implementation of different scheduling algorithms using Shell script programs. Comparing types of process scheduling algorithms in terms of minimum Turn-around time, Waiting time and response time. Implement The Producer / Consumer Problem, Banker's algorithm in Linux/Unix shell scripts.		
3	Memory Management			
	3.1	Basic concept, Logical and Physical address maps, Memory allocation strategies fragmentation and Compaction.	06	CO3
	3.2	Basics of Virtual Memory, Page allocation, Partitioning, Paging, Page fault, Segmentation, Demand paging, Page Replacement algorithms: Optimal, First in First Out (FIFO) and Least Recently used (LRU).		

4	File Management		06	CO4
	4.1	Concept of File, Access methods, File types, File operation, Directory structure, File System structure.		
	4.2	File allocation methods (contiguous, linked, indexed), Free-space management (bit vector, linked list, grouping), directory implementation. #Self-Study: Implementation of File allocation methods using Linux/Unix shell scripts.		
5	Storage and I/O management		04	CO5
	5.1	I/O Hardware: I/O devices, Device controllers, Direct Memory Access, Principles of I/O. Disk Management: Disk structure, Disk scheduling - FCFS, SSTF, SCAN, C-SCAN.		
	5.2	#Self-Study: Implementation of File allocation methods using Linux/Unix shell scripts.		
		Total	30	

- LAB CA includes journal / written record of experiments based on above contents.
- LAB CA also includes evaluations covering entire syllabus
- LAB CA schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge.

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Silberschatz A., Galvin P., and Gagne G	<i>Operating Systems Concepts</i>	Wiley Eastern Limited, India	9 th Edition, 2018
2	William Stallings,	<i>Operating System- Internal & Design Principles</i>	Pearson, London, England	7 th Edition, 2013
3	Maurice J. Bach	<i>The Design of Unix Operating System</i>	Prentice Hall, United States	1 st Edition 2015
4	Andrew S. Tanenbaum. Herbert Bos	<i>Modern Operating Systems</i>	PHI Learning Pvt. Ltd, India	4 th Edition 2016

ELECTIVE –I LABORATORY

Course Code	Name of the Course				
216U03L511	Random Processes Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
		50	--	--	--

*Distribution of marks:

Instructions:	- LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8 experiments will be performed in a semester. - LAB CA also includes evaluations covering entire syllabus (50 marks). - LAB CA schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge.
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Course Code	Name of the Course				
116U03L512	Control Systems and Applications Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
		50	--	--	--

Instructions:	- LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8 and maximum 10 experiments will be performed in a semester. - LAB CA also includes evaluations covering entire syllabus (50 marks). - LAB CA schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge.
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Course Code	Name of the Course				
216U03L513	Database Management System Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

*Distribution of marks:

Instructions:	- LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8-10 experiments will be performed in a semester. - LAB CA also included Evaluations based on entire syllabus (25 marks). - LAB CA Evaluations l schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge.
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Course Code	Name of the Course				
216U03L514	Soft Computing Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

*Distribution of marks:

Instructions:	• LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8-10 experiments will be performed in a semester. • LAB CA also included Evaluations based on entire syllabus (25 marks). • LAB CA Evaluations schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge.
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Course Code	Name of the Course				
216U03L515	Machine Learning Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
		50	--	--	--

*Distribution of marks:

Instructions:	- LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8-10 experiments will be performed in a semester. - LAB CA also included Evaluations based on entire syllabus (25 marks). - LAB CA Evaluations schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge.
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Course Code	Name of the Course				
216U03P501	Mini Project				
Teaching Scheme	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
		50	--	--	--

Course Objectives: The Mini Project course is designed to enable students to apply the theoretical knowledge and technical skills acquired from previous courses as well as relevant to field to identify and implement solutions for real-world problems. The course aims to apply fundamentals of project management, emphasizing planning, execution, time management, and cause-effect analysis among various project dimensions. It encourages innovation and creativity by allowing projects that go beyond the standard curriculum, thereby promoting learning of additional tools, technologies, and design methodologies along with team collaboration, leadership, and interpersonal skills through group-based project execution. Working on real world applications also imbibes a sense of social responsibility. The solutions of problem statements aligned with domains such as Agriculture, Healthcare, Automation & Artificial Intelligence, Embedded systems, Energy and Environment etc are encouraged.

Course Outcomes (CO):

Upon successful completion of the Mini Project course, students will be able to:

- CO1.** Identify and define a relevant engineering problem, clearly stating its objectives and scope.
- CO2.** Analyze system requirements and design appropriate components or modules to meet the desired specifications.
- CO3.** Implement, test, and validate the proposed solution using suitable tools, technologies, and methodologies.
- CO4.** Collaborate effectively within a team, adhering to timelines, demonstrating responsibility, and applying principles of behavioural science.
- CO5.** Communicate project ideas, processes, and outcomes effectively through written reports and oral presentations, while adhering to ethical and professional standards.

Evaluation of CA:

- The mini project will be evaluated on continuous basis and will be graded accordingly.
- CA Evaluations schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge.



SEMESTER VI

Course Code	Name of the Course				
216U03C601	Wireless Communication				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites:

- Communication Systems
- Digital Communications

Course Objectives: The objective of this course is to impart knowledge of evolution of wireless communication and fundamental radio concepts, which are at the core of providing mobile communication service to subscribers on the move using limited radio spectrum. Radio propagation losses both large and small describes how to measure and model the impact that signal bandwidth and motion have on the instantaneous received signal through a multipath channel. Second and third generation systems like GSM, CDMA IS95, UMTS are covered. OFDM, MIMO and long-term evolution systems are introduced.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Comprehend fundamentals of mobile communication
- CO2.** Evaluate various path loss and fading effects
- CO3.** Compare principles of GSM and IS-95 system.
- CO4.** Illustrate basics of 3G UMTS technology.
- CO5.** Summarize enabling technologies for 3GPP LTE-A network

Detailed Syllabus

Module No.	Unit No.	Details	Hrs	CO
1	Fundamentals of Mobile Communication		10	CO1
	1.1	Telecom Regulatory Authority of India (TRAI) act & policies		
	1.2	Frequency reuse, and Channel assignment strategies.		
	1.3	Handoff strategies, Practical handoff considerations, Co-channel interference and system capacity, Adjacent channel interference.		
	1.4	Trunking, Grade of Service and related design problems.		
	1.5	Improving Coverage and Capacity in Cellular System: Cell splitting, Cell sectoring, A microcell zone concept and Repeaters for range extension.		
2	Mobile Radio Propagation		08	CO2
	2.1	Large Scale Path Loss: The three basic propagation mechanisms: Reflection, Diffraction, and Scattering.		
	2.2	Small Scale Fading and Multipath: Factors influencing small scale fading, and Doppler shift		
	2.3	Parameters of mobile multipath channels: Time dispersion parameter, Coherence bandwidth, Dopplerspread and coherence time and related numerical.		
	2.4	Types of small scale fading: Fading effects due to multipath time delay spread, Fading effects due to Doppler spread.		
		#Self-learning topic: Outdoor Propagation Models: Longley-Rice Model, Okumura Model, Hata Model Indoor Propagation Models: Log-distance path loss model, Attenuation factor model.		
3	2G/2.5G Technologies		11	CO3
	3.1	Global System for Mobile Communication Standard (GSM): GSM features, services, architecture, channel types, call processing, frame structure and signal processing		
	3.2	GSM evolution: GPRS features and network architecture		
	3.3	Interim Standard-95 (IS-95): Air interface specifications, channel types, call processing, power control.		
		#Self-learning topic: Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA), Direct Sequence Spread Spectrum (DSSS)		
4	3G Technologies		04	CO4
	4.1	Universal Mobile Telecommunication Standard (UMTS): The WCDMA Air Interface, Objectives, network architecture, UTRAN architecture of UMTS		
	4.2	UMTS Channel types and functions		
	4.3	Codes used in UMTS, Spreading and Scrambling in UMTS		

5	Enabling Technologies for 3GPP LTE-Advanced Networks			
	5.1	Orthogonal Frequency Division Multiplexing (OFDM): Basic principle of orthogonality, single carrier vs. multicarrier systems, block diagram, cyclic prefix, and Peak to average Power ratio.	12	CO5
	5.2	Long Term Evolution (LTE): Air interface specifications, Architecture and LTE physical layer		
	5.3	LTE-A Enabling Technologies: Carrier aggregation, relaying, and Self-optimizing networks (SON).		
	5.4	Concept of MIMO, spatial diversity, spatial multiplexing, antenna beamforming and space time block codes		
	5.5	Cognitive Radio Networks: Definition and Spectrum sensing		
	5.6	Introduction to 5G network, specifications, applications, use cases, Comparison of 5G and 4G network and applications		
		Total	45	

Recommended Books*:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Theodore S. Rappaport	<i>Wireless Communications Principles and Practice</i>	Pearson Publication, India	2 nd Edition, 2010
2.	T L Singal	<i>Wireless Communications</i>	Tata McGraw Hill, India	2 nd Edition, 2017
3.	Guillaume de la Roche, Ben Allen	<i>LTE-Advanced and Next Generation Wireless Networks-channel modelling and propagation</i>	Wiley Publication, India	1 st Edition, 2013
4.	Young Kyun Kim and Ramjee Prasad	<i>4G Roadmap and Emerging Communication Technologies</i>	Artech House Publishers, Boston, London	1 st Edition, 2005
5.	Aditya K. Jagannathan	<i>Principles of Modern Wireless Communication Systems: Theory and Practice</i>	McGraw-Hill Education, India	1 st Edition, 2017
6.	Upena Dalal	<i>Wireless Communication</i>	Oxford University Press, India	2 nd Edition, 2010

Course Code	Name of the Course				
216U03C602	RF Filters and Antennas				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites:

- Electromagnetic Field theory
- Electrical Networks

Course Objectives:

The objective of this course is to enable the students to understand and analyze radio frequency behavior of transmission line and passive components like resistor, inductor and capacitor, design and analysis of filters by insertion loss method. The course covers the analysis of wire antennas, antenna arrays and special type of antennas like microstrip, helical, reflector needed for the communication system

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Evaluate transmission lines parameters analytically and using smith charts.
- CO2.** Evaluate the high frequency behaviour of passive components and design different types of filters.
- CO3.** Understand the radiation characteristics of different wire antennas.
- CO4.** Analyse the radiation pattern of array.
- CO5.** Describe special type of antennas and their applications.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs	CO
1	Introduction		12	CO1
	1.1	Introduction to RF, RF spectrum and applications, two wire transmission line and its equivalent circuit representation and transmission line parameters, types of transmission lines: Coaxial transmission line, parallel plate and microstrip transmission line.		
	1.2	Terminated lossless transmission lines, special termination conditions, source and loaded transmission line.		
	1.3	Introduction to smith chart, admittance transformation, transformation from reflection coefficient to load impedance using smith chart.		
	1.4	Impedance transformation using smith chart. #Self-Learning:RF hazards		
2	RF Passive Components and Filter Design		08	CO2
	2.1	RF behavior of passive components: Resistors, capacitors & inductors, introduction to chip components.		
	2.2	Filter parameters and configuration, Butterworth and Chebyshev type filter realization of low pass, high pass, band pass and band reject using insertion loss method for equal source and load resistance		
	2.3	Filter transformation, Impedance and frequency scaling, Microstrip filters design, Richard's transformation and Kuroda's identity. (Low pass filter design) Self Learning: S parameters		
3	Fundamental Parameters of Antennas		09	CO3
	3.1	Introduction to radiation mechanism, current distribution on thin wire antenna.		
	3.2	Antenna parameters: Radiation pattern, solid angle, radiation power density, radiation intensity, beamwidth, directivity, gain, antenna efficiency, beam area, beam efficiency, bandwidth , aperture concept, polarization, input impedance, antenna temperature, near and far field, Maximum directivity and maximum effective area, Friis transmission equation		
	3.3	Linear wire antennas: Potential functions, infinitesimal dipole, finite length dipole and half wave length dipole. (No derivations for Electric and Magnetic Field of wire antennas) #Self-Learning: Radiation integrals and auxiliary potential functions		

4	Arrays		
	4.1	Introduction to antenna array, types and controlling parameters of array	08 CO4
	4.2	Array of two isotropic point sources, non-isotropic sources and linear arrays of n elements: broadside and endfire array.	
	4.3	Principle of pattern multiplication and plot radiation pattern of an array.	
5	Special Types of Antennas		
	5.1	Application of linear wire antennas: Monopole antenna, Yagi Uda antennas design and applications	08 CO5
	5.2	Helical antennas: Modes of operation and design of axial mode helix.	
	5.3	Reflector antennas: Plane reflector, parabolic reflector and its feed mechanism.	
	5.4	Microstrip antennas: Rectangular microstrip antenna (RMSA) structure, transmission line model, advantages, disadvantages, applications and feed mechanism, design of RMSA.	
		Total	45

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Reinhold Ludwig, Pavel Bretchko	<i>RF Circuit Design</i>	Pearson Education Asia, United States	2nd Edition, 2013
2	C. A. Balanis	<i>Antenna Theory Analysis and Design</i>	John Wiley & Sons Inc, India	3 rd Edition, 2005
3	John D. Kraus Ronald J. Marhefka, Ahmad S. Khan	<i>Antenna and wave Propagation</i>	Tata McGraw Hill publication, India	5 th Edition, 2017

Course Code	Name of the Course				
216U03C603	Optical Fiber Communication				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites:

Solid State Physics, Optics, Wave Theory

Course Objectives:

The objective of this course is to understand the fundamental principles of optical fiber technology, optical components and their interactions, behaviour of basic digital/analog systems, analyse the performance characteristics and applications in modern telecommunication systems.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Apply the principles of optics to optical fiber wave propagation and understand fiber Fabrication techniques
- CO2.** Understand the transmission characteristics and signal degradation in optical fiber Communication.
- CO3.** Analyse optical sources and detectors, Power launching and coupling efficiencies.
- CO4.** Analyse the working of optical Receiver, Digital and Analog systems transmission systems.
- CO5.** Analyze the optical fiber systems and components required for optical communication Link.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	Optical Fibers: Structures, Waveguiding and Fabrication		07	CO1
	1.1	Optical fiber Structures, Wave guiding and Fabrication: Optical Spectral bands, Basic optical laws and definitions.		
	1.2	Optical fibers Modes and configurations, Single mode fibers, Multi-mode fiber Graded index fiber structures,		
	1.3	Mode theory for circular waveguide, LP modes and power flow.		
	1.4	Fiber materials, fiber fabrication, Mechanical properties.		
2	Signal Degradation in Optical Fibers		12	CO 2
	2.1	Signal Degradation in optical fiber: Attenuation, bending losses, Scattering, Absorption, core cladding losses.		
	2.2	Signal distortion: Dispersion in fibers, limitations due to dispersion, Material, waveguide, polarization dispersion, Intermodal dispersion.		
	2.3	Design optimization of single mode fiber: Dispersion shifted and Dispersion flattened fibers, RI optimization, MFD. bandwidth-length product Non-Linear losses in optical fibers		
3	Optical Sources, Detectors, Power launching and coupling		10	CO3
	3.1	Optical Sources and Detectors : Direct and Indirect band gap materials.		
	3.2	Light Emitting (LED's): Structures, quantum efficiency and power, Modulation. Laser Diodes: Laser diode rate equations, Structures and radiation patterns, Single mode lasers, efficiency.		
	3.2	Properties of photo diodes, structures, PIN and APD, photo detector noises, Avalanche Multiplication Noise, Response time, comparison of photodiodes.		
	3.3	Power launching, and coupling, fiber to fiber joints and losses, splicing types, connectors and types.		
4	Optical Receiver, Digital and Analog systems		08	CO4
	4.1	Optical Receiver Operation, receiver structures, pre-amplifier types, Noises and performance calculations.		
	4.2	Point to point links, Digital receiver performance, Link power budget, Rise time budget, transmission distance and noise effects.		
	4.3	Coherent detection Optical detection, Relative intensity noise, Multichannel transmission. amplifiers, basic applications and types, semiconductor optical amplifiers, EDFA		

5	Optical Fiber Systems and components			
	5.1	Optical amplifiers, basic applications and types, semiconductor optical amplifiers, EDFA, wavelength converters.	08	CO5
	5.2	Operational Principles of WDM and DWDM, Standards, Optical Isolators and Circulators, Fiber Grating Filters, Optical Add/Drop Multiplexing, Optical switching networks.		
	5.3	Optical system parameter measurements using OTDR, Eye patterns		
Total			45	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Gerd Keiser	<i>Optical Fiber</i>	<i>Mc- Graw Hill Publication, India</i>	4 th Edition, 2011
2	John Senior	<i>Optical Fiber Communication</i>	<i>Prentice Hall of India Publication</i>	4 th Edition, 2014

Course Code	Name of the Course				
216U03E611	Basic VLSI Design				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100
Course pre-requisites: Concepts of Electronic Devices, Digital Circuits, Integrated circuits					
Course Objectives: This course provides students basic knowledge and skills in the fundamental theories and practices of Cyber Security. The course will gives the knowledge and techniques of protecting computers, networks, programs and data from unauthorized access or attacks that are aimed for exploitation.					
Course Outcomes (CO): At the end of successful completion of the course the student will be able to CO1. Demonstrate an understanding of MOS fundamentals CO2. Design and analyse combinational logic circuits CO3. Design and analyse sequential logic circuits CO4. Implement different dynamic and low power CMOS circuit CO5. Implementation Digital circuits using HDL in CPLD/FPGA					

Detailed Syllabus

Module No.	Unit No.	Details	Hrs.	CO
1	Introduction		08	CO 1
	1.1	Introduction, Revolutions in VLSI, Overview of VLSI design flow		
	1.2	Semiconductor physics of MOSFET, MOSFET Types: PMOS, NMOS and CMOS; threshold voltage calculations ;MOSFET capacitances, MOSFET Fabrication		
	1.3	Stick diagram and Layout design (using lamda rules), scaling		
	1.4	Self-Study: NMOS, PMOS and CMOS fabrication process flow and Basic processes		
2	MOSFET combinational circuits		08	CO2
	2.1	Inverter Characteristics, CMOS Inverter Operation, Voltage Transfer Characteristics, Noise Margins and Power consumption		
	2.2	Transmission Gate (TG), Pass Transistor Logic (PTL), Complementary pass transistor Logic (CPL), CPL based logic dates and combinational circuits		
	2.3	Logic circuit Design: Analysis of two input NOR and NAND gate using equivalent CMOS Inverter		
	2.4	Arithmetic building blocks: Adder, ripple carry adder, carry look ahead adder, Multipliers: Array multiplier, Wallace Tree multiplier		
3	Sequential MOS Logic Circuits		08	CO3
	3.1	Bistable circuits, Design for sequential circuits like SR latch, Clocked latch and Flip-Flops , J-K flip flop, D latch and edge triggered Flip-Flop, Sense-Amplifier-Based Flip-Flops, Power consumption of clocking system and power saving methodologies		
	3.3	Semiconductor Memories, memory types, Dynamic RAM, configuration of DRAM, DRAM cell types, Flash Memory		
4	Dynamic Logic Circuits and Low power CMOS circuits		10	CO4
	4.1	Basic Principle of pass Transistor circuits, Voltage Bootstrapping		
	4.2	Synchronous Dynamic circuit techniques, Dynamic Pass Transistor circuits, Dynamic CMOS circuits techniques,		
	4.3	High-Performance Dynamic CMOS Circuits, Domino CMOS Logic, NORA CMOS Logic, Zipper CMOS Logic, TSPC Dynamic CMOS		

	4.4	Overview of power consumption, Low power design through voltage scaling, variable-threshold CMOS circuits, Parallel processing approach		
5	Programmable logic Devices and Hardware Description Languages		10	CO5
	5.1	Architecture of PLDs, PROM (Programmable Read-Only Memory), PAL (Programmable Array Logic), PLA (Programmable Logic Array), Complex PLDs (CPLDs), Field-Programmable Gate Arrays (FPGAs), Implementation of digital circuits in CPLD and FPGA		
	5.2	Types of HDLs, Basic constructs and modelling using VHDL (Very High-Speed Integrated Circuit Hardware Description Language) and Verilog HDL		
	5.3	Digital Circuits Using HDL, Combinational Circuits: Multiplexers, Encoders, Decoders, MUX in Verilog and VHDL. Sequential Circuits: Flip-Flops (D, JK, T), Counters and Shift Registers		
Total			45	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Sung-Mo Kang, Yusuf Leblebici, Chulwoo Kim	<i>CMOS Digital Integrated Circuits</i>	McGraw-Hill Education	4 TH Edition, 2015
2	John P. Uyemura	<i>VLSI Circuits and Systems</i>	Wiley Publications, India	1st Edition, 2001
3	Neil H. E. Weste, Kamran Esheaghian	<i>Principles of CMOS VLSI Design</i>	Pearson Education India,	4th Edition, 2006
4	J. Bhaskar	<i>A VHDL Primer</i>	Pearson	3rd Edition, 2020
5	Samir Palnitkar	<i>Verilog HDL A guide to Digital Design and Synthesis</i>	SunSoft Press	2 nd Edition 1996

Course Code	Name of the Course				
216U03E612	Instrumentation System				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course prerequisites:

- Analog Electronics., Basic Electronics, Electrical Networks

Course Objectives:

Objectives of this course are:

- To understand the fundamental concepts of Electronics Measuring instruments.
- To explore the functioning and applications of various transducers such as temperature, displacement
- To learn Concepts of signal conditioning and data acquisition circuits
- To do case study for any feedback system through activity based learning.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Understand the working principle of various laboratory instruments
CO2. Choose a transducer for given application
CO3. Choose an appropriate signal conditioning circuits for instrumentation purpose
CO4. Rig up a circuit for measuring various physical parameters
CO5. Explore the working of PID controllers and PLC

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	Introduction to Measurement and Instruments		08	CO1
	1.1	Fundamentals of Measurement: Units, Standards, and Measurement Errors		
	1.2	Electrical Measurement Instruments: Analog and Digital Instruments (Voltmeter, Ammeter, Wattmeter, Energy Meter)		
	1.3	Digital Instruments: Digital Storage Oscilloscopes (DSO), Arbitrary Function Generators, Spectrum Analyzers		
2	Sensors and Transducers		08	CO2
	2.1	Classification and Selection of Sensors and Transducers: Strain Gauges, Inductive & Capacitive Sensors, Piezoelectric, Hall Effect Transducers		
	2.2	Thermal Sensors (Thermistors, Thermocouples), Optical Sensors (Photodiodes, Phototransistors), Digital Transducers (Encoder Types)		
	2.3	Instrument Transformers: Current and Potential Transformers, Ratio and Phase Angle Errors		
3	Signal Conditioning Circuits		08	CO3
	3.1	Instrumentation Amplifiers, Precision Rectifiers, Isolation Amplifiers, Log Amplifiers, Charge Amplifiers		
	3.2	Noise Reduction Techniques in Instrumentation, Basics of Smart Sensors, Data Acquisition Systems (DAQ), Signal Conditioning and Telemetry		
4	Measurement of various Elements		07	CO4
	4.1	Displacement Measurement: Resistive, Capacitive, and Inductive Sensors, Differential Pressure Measurement		
	4.2	Torque Measurement (Strain Gauge Methods), Flow Measurement (Hotwire Anemometer, Eddy Current Sensors)		
	4.3	Phase Measurement: Analog and Digital Phase Detectors		
5	Process Control And Automation		14	CO5
	5.1	Basic Control Actions: ON-OFF, Proportional, Integral, Derivative Controls; PID Controllers; Controller Tuning Techniques (Ziegler-Nichols, Process Reaction Curve, etc.)		
	5.2	Programmable logic controllers, PLC architecture, programming, Relay ladder logic, Introduction to Sequence Control, PLCs - Working, Specifications, concept of latching, Introduction to Timer/Counters, Exercises based on Timers, Counters.		
	5.3	Advanced Process Automation: Introduction to SCADA, DCS, and IIoT Applications in Instrumentation		
Total			45	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	A. K. Sawhney	<i>Electrical and Electronics Measurements and Instrumentation</i>	Dhanpat Rai and son's Ltd, New Delhi	2 nd Edition, 2003
2	DVS Murthy	<i>Transducers and Instrumentation</i>	PHI, Learning Pvt Ltd New Delhi	2 nd Edition, 2012
3	Curtis D Johnson	<i>Process Control Instrumentation Technology</i>	PHI, Learning Pvt Ltd New Delhi	8 th Edition, 2013
4	Doebelin E.O	<i>Measurement Systems: Application and Design</i>	McGraw Hill, Newyork,	4 th Edition, 1992
5	Madhuchhanda Mitra, Samarjit Sengupta	<i>Programmable Logic Controllers and Industrial Automation An Introduction</i>	Penram International Publishing (India) Pvt Ltd.,	2 nd Edition, 2017
6	S. Nihtianov, A. Luque	<i>Smart Sensors and MEMS: Intelligent Sensing Devices and Microsystems</i>	Woodhead Publishing	2nd Edition, 2018

Course Code	Name of the Course				
216U03E613	Digital Image Processing				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites:

- Signals and Systems
- Digital Signal Processing
- Basics of Matrices

Course Objectives:

Digital Image Processing is one of the most widely investigated subjects under the domain of signal processing. The objective of the course is to introduce students the fundamentals of images and their processing using various techniques. The topics covered in this course find wide scope in applications such as photographic image processing, satellite imaging, bio-medical image processing, video codecs, etc. The course also aims to prepare students for further in-depth study of this field.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Understand theory and models in Image Processing
- CO2.** Interpret and analyze image enhancement algorithms in spatial domain
- CO3.** Interpret and analyze 2D signals in frequency domain through image transforms
- CO4.** Apply various methods for image segmentation and image description
- CO5.** Develop and design practical applications in various fields

Detailed Syllabus

Module No.	Unit No.	Details	Hrs	CO
1	Image Fundamentals		06	CO1
	1.1	Image acquisition, sampling and quantization, Spatial and Tonal resolution of images, types and formats of images, basic relationship between pixels, basic operations between images such as image addition, subtraction, blending and convolution. Color images: RGB and HSI models		
	1.2	# Self-Learning topics: Case study: Image fusion		
2	Image Enhancement in Spatial Domain		10	CO2
	2.1	Point Processing methods: Digital Negative, contrast stretching, thresholding, gray level slicing, bit plane slicing and log transform		
	2.2	Neighborhood Processing: Averaging filters, order statistics filters, high pass filters and high boost filters and their applications		
	2.3	Image Histogram stretching and equalization		
	2.4	Introduction to Image steganography, case studies on Processing DICOM files and Medical Imaging such as X ray, CT, MRI		
	2.5	# Self-Learning topics: Image zooming		
3	Frequency Domain Processing		10	CO3
	3.1	Two Dimensional Transforms: Concept of basis images, Discrete Fourier Transform (DFT), properties of DFT such as separability, frequency shift, average value and convolution, Discrete Cosine Transform, Hadamard Transform, Haar Transform		
	3.2	Frequency Domain filtering: DFT for filtering, Ideal, Gaussian, Butterworth filters for smoothening and sharpening, and Homomorphic filters.		
	3.3	Image Restoration: Degradation model, noise models, restoration using Wiener filters and Inverse filters		
		# Self learning topic: Walsh-Hadamard and KL Transform		
4	Image Segmentation, Description and Morphology		11	CO4
	4.1	Segmentation based on discontinuities: Point, line and edge detection.		
	4.2	Segmentation based on similarities and region based segmentation: Region growing, region splitting, region merging, split and merge		
	4.3	Boundary description: Chain codes, Shape number, Euler's number, Signatures, Fourier descriptors and statistical moments up to first order		

	4.4	Morphology: Dilation, erosion, opening, closing, hit or miss transform, thinning and thickening, and boundary extraction on binary images		
		# Self learning topic: Edge linking using Hough transforms.		
5	Image Compression		08	CO5
	5.1	Fundamentals of image compression and error criteria, definition of compression ratio, Mean Squared Error (MSE) and Peak Signal to Noise Ratio (PSNR)		
	5.2	Lossless Compression Techniques: Run length Encoding, Huffman codes, LZW technique		
	5.3	Lossy Compression Techniques: Improved Grey Scale Quantization, Transform coding		
		# Self learning topic: JPEG compression standard		
Total			45	

Reference Books*

Sr. No.	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1.	Jayaraman, Esakkirajan and Veerakumar	<i>Digital Image Processing</i>	Tata McGraw Hill Education Pvt. Ltd., India	2 nd reprint, 2010
2.	Gonzalez and Woods	<i>Digital Image Processing</i>	Pearson Education, India	3 rd Edition, 2008
3.	Stan Birchfield	<i>Image Processing and Analysis</i>	Cenage Learning, USA	1 st Edition 2016
4.	Oge Marques	<i>Practical Image and Video Processing using MATLAB</i>	John Wiley & Sons Inc., US	1 st Edition, 2011

Learners should prepare all self-learning topics on their own. Self-learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in IA.

* No questions expected on case study in ISE, IA and end semester theory examination.

Course Code	Name of the Course				
216U03E614	Cyber Security				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Computer networks

Course Objectives: This course provides students basic knowledge and skills in the fundamental theories and practices of Cyber Security. The course will give the knowledge and techniques of protecting computers, networks, programs and data from unauthorized access or attacks that are aimed for exploitation. The course also provides an overview on the future trends in the security domain.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Understand the broad set of technical, social & political aspects of Cyber Security
- CO2.** Understand the nature of secure software development and operating systems
- CO3.** Discuss the importance of attacks and preventive measures at network and application layer
- CO4.** Discuss strategies for cyber defense and incident response
- CO5.** Elaborate about cyber security policies, risk management and future trends in security domain.

Detailed Syllabus

Module No.	Unit No.	Details	Hrs.	CO
1	Introduction to cyber security		07	CO1
	1.1	Introduction to Cyber Security and Information Security		
	1.2	Threats , harms, vulnerabilities and controls		
	1.3	Attack Vectors (Types of Attacks, Types of Malwares – Worms, Phishing, Trojans, Botnets, Root Kits and Viruses etc),Types of security, goals of security		
	1.4	Need of Cryptography(public and private key) hashing, session keys		
2	Software and System Security		13	CO2
	2.1	Firewalls Basics, Stateless packet filtering, Stateful/Session Filtering, Application Level Firewalls, Circuit Level Gateways, Packet Filter Vs Firewall, Packet Characteristic to Filter, Linux Firewall, Windows Firewall, Network Address Translation (NAT) and Port Forwarding, Basics of Virtual Private Networks		
	2.2	Security in Operating Systems, Access Control and Authentication, Patch Management and Updates, Secure Configuration and Hardening		
	2.3	Secure Software Development Lifecycle (SDLC)		
3	Network Security & Application Security		11	CO3
	3.1	Security Issues in TCP/IP- TCP, DNS, Routing Protocol (Routing Security –BGP ,vulnerabilities, BGP Security etc)		
	3.2	Web Application Security-OWASP security and Large Language Model Applications security		
	3.3	Content Security Policies (CSP) in web – HTTP Header Directives and Sources, Session Management and User Authentication, Session Integrity, HTTPS, SSL/TLS (working principle)		
4	Cyber Threats and Defense Mechanism		08	CO4
	4.1	Denial of Service (DoS) and Distributed Denial of Service (DDoS) Attacks, Advanced Persistent Threats (APTs)		
	4.2	Cyber Defense Strategies ,Intrusion Detection System/ Intrusion Prevention System – Overview, Anomaly Based Detection Methods, Signature Based IDS/IPS, Network based IDS/IPS,		

		Host Based IDS/IPS, SNORT, Honeypot		
	4.3	Challenges ,Best Practices, Security Audits and Assessments, Limitations ,Techniques to visualize network data		
5	Cyber Security Policies, Risk Management and Future Trends		06	CO5
	5.1	Cyber Security Governance and Policies, Risk Management Frameworks, Digital and Cyber Forensics Compliance and Legal Issues in Cyber Security, Business Continuity and Disaster Recovery Planning, Privacy and Data Protection Regulations		
	5.2	Overview of Industrial Internet of Things (IIoT) Security, Cloud Security, Artificial Intelligence and Cyber Security ,Quantum Cryptography		
Total			45	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Nina Godbole, Sunit Belapure	<i>Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives</i>	Wiley india Pvt. Ltd, India	1st edition copyright 2011 reprint 2013
2	<u>Mike Shema</u>	<i>Anti-Hacker-Tool -Kit</i>	McgrawHill, India	4th Edition February 2014
3	Charles Pfleeger, Shari Lawrence Pfleeger, Jonathan Margulies	<i>Security in Computing</i>	Prentice Hall, India	5th Edition February 5, 2015
4	William Stallings	<i>Network Security Essentials: Applications and Standards</i>	Pearson	February ,2018
5	Chwan-Hwa (John)Wu J. David Irwin	<i>Computer Networks and Cyber Security</i>	CRC Press	First Indian Reprint, 2014

Course Code	Name of the Course				
216U03E615	Cloud Computing				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Computer communication networks
Network security

Course Objectives: Cloud computing services are being adopted widely across a variety of organizations and in many domains. This course will introduce the fundamental concept of cloud computing, it's enabling technologies, virtualization and security concerns. This course also imparts the functional capabilities of popular cloud platforms such as GAE, AWS and Azure.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Understand different cloud service models

CO2. Differentiate between various virtualization techniques

CO3. Recognize different cloud programming structures and usage from business perspective

CO4. Understand several security mechanisms for cloud environment

CO5. Understand challenges of maintaining security in cloud architecture

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs	CO
1	Cloud Computing fundamentals		08	CO1
	1.1	Overview of Distributed Computing, Cluster computing, Grid computing.		
	1.2	Evolution, characteristics, and components of cloud computing, Advantages of Cloud computing		
	1.3	Cloud service Models, Cloud Deployment Models , Alternative Deployment Models		
	1.4	Relevant Deployment Models for Enterprise Cloud Computing		
	1.5	Centralization of Data Storage		
2	Virtualization techniques		10	CO2
	2.1	Virtual Machines and Virtualization of Clusters and Data Centers, Virtualization Structures/Tools and Mechanisms		
	2.2	Implementation Levels of Virtualization, Server Virtualization: Understanding Server Virtualization, types of server virtualization		
	2.3	Virtualization Structures/Tools and Mechanisms, Virtualization of CPU, Memory and I/O Devices		
	2.4	Virtual Clusters and Resource Management, Virtualization of Data-Center Automation		
3	Cloud computing architectures and Data Centers		09	CO3
	3.1	Data-Center design and Interconnection networks		
	3.2	Architectural Design of Compute and Storage Clouds		
	3.3	Virtual machine migration and Load balancing		
	3.4	Web services - Web 2.0 - Web OS - Case studies – Anything as a service (XaaS).		
4	Cloud Platforms: overview and services		10	CO4
	4.1	Programming support of Google App Engine: Programming the Google App Engine, Google file system(GFS), Bigtable, Googles NOSQL system. Chubby, Google’s Distributed lock service		
	4.2	Programming on Amazons AWS: Programming on Amazon EC2, Amazon simple storage service(S3), Amazon Elastic block store(EBS), and SimpleDB, Virtual private cloud		
	4.3	Programming on Microsoft Azure: Core architecture components of Azure, Azure compute, networking, storage services		

5	Security architecture and Challenges		08	CO5
	5.1	Architectural Considerations- General Issues, Trusted Cloud Computing, Secure Execution Environments and Communications, Micro-architectures; Identity Management and Access control-Identity management, Access control		
	5.2	Virtualization Security, VPN, Management Virtual Threats , Hypervisor Risks ,Increased Denial, of Service Risk , VM Security Recommendations		
	5.3	Cloud federation,: EU cyber solidarity act, interoperability and standards: ISO 27018		
Total			45	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	KaiHwang, Geoffrey C. Fox, Jack J. Dongarra,	<i>Distributed And Cloud Computing From parallel processing to the internet of things</i>	ELSEVIE MK publishers	First, 2012
2	Ronald L. Krutz, Russell Dean Vines	<i>Cloud Security A Comprehensive Guide to Secure Cloud Computing</i>	Wily	First, 2010
3	Antohy T Velte	<i>Cloud Computing : A Practical Approach</i>	McGraw Hill	10 th Edition 2004
4	Rajkumar Buyya, James Broberg, Andrzej Goscinski	<i>Cloud Computing, Principles and Paradigms</i>	Wiley	First ,2013

Course Code	Name of the Course				
216U03L601	Wireless Communication Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

*Distribution of marks:

Instructions:	- LAB CA includes journal / written record of experiments. A minimum of 8-10 experiments will be performed in a semester based on the syllabus of wireless communication 216U03C601. - LAB CA also includes Evaluation based on entire syllabus/experiments. - LAB CA schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge(s).
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Course Code	Name of the Course				
216U03L602	RF Filters and Antennas Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

*Distribution of marks:

Instructions:	- LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8-10 experiments will be performed in a semester. - LAB CA also includes Evaluation based on entire syllabus/experiments. - LAB CA schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge(s).
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Course Code	Name of the Course				
216U03L603	Optical Fiber Communication Labotatory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

Instructions:	- LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8-10 experiments will be performed in a semester. - LAB CA also includes Evaluation based on entire syllabus/experiments. - LAB CA schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge(s).
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Course Code	Name of the Course			
216U03L611	Basic VLSI Design Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01
Evaluation Scheme	Marks			
	LAB/TUT CA*	CA (TH)		ESE
		IA	ISE	
	50	--	--	50

*Distribution of marks:

Instructions:	- LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8-10 experiments will be performed in a semester. - LAB CA also includes Evaluation based on entire syllabus/experiments. - LAB CA schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge(s).
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Course Code	Name of the Course				
216U03E5612	Instrumentation System Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

*Distribution of marks:

Instructions:	• LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8-10 experiments will be performed in a semester. • LAB CA also includes Evaluation based on entire syllabus/experiments. • LAB CA schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge(s).
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Course Code	Name of the Course				
216U03E613	Digital Image Processing Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

*Distribution of marks:

Instructions:	- LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8 and maximum 10 experiments will be performed in a semester. - LAB CA also includes evaluations covering entire syllabus (50 marks). - LAB CA schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge. The total marks for LAB CA will be 50.
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Course Code	Name of the Course				
216U03E614	Cyber Security Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
		50	--	--	--

Instructions:	- LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8-10 experiments will be performed in a semester. - LAB CA also includes Evaluation based on entire syllabus/experiments. - LAB CA schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge(s).
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Course Code	Name of the Course				
216U03E615	Cloud Computing Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
		50	--	--	--

*Distribution of marks:

Instructions:	- LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8-10 experiments will be performed in a semester. - LAB CA also includes Evaluation based on entire syllabus/experiments. - LAB CA schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge(s).
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Course Code	Name of the Course				
216U03P601	Project-I				
Teaching Scheme	TH	P	TUT	Total	
	--	04	--	04	
Credits Assigned	--	02	--	02	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

Course Objectives: The objective of Major Project – Stage I is to provide students an opportunity to undertake a comprehensive engineering project in a group, applying theoretical knowledge and practical skills to solve real-world or research-based problems. Encourage students to pursue innovative solutions by extending previous mini project work or initiating a new, technically challenging idea relevant to their academic or career interests.

Students are expected to follow the process of defining, designing, analyzing, and planning for implementation of a technical solution, using both hardware and software tools e.g. simulation, modeling, data analysis, prototyping etc.

The project facilitates development of teamwork, leadership, and project management skills through regular progress reviews, reporting, and collaboration with faculty advisors. Students can work on problem statements in the areas like Agriculture Technology, Healthcare Systems, Automation and Artificial Intelligence, Information and Cyber Security, Energy Efficiency and Environmental Solutions etc.

Course Outcomes (CO):

Upon successful completion of Major Project – Stage I, students will be able to:

- CO1.** Identify and formulate a real-world engineering problem, with clear objectives, scope, and constraints.
- CO2.** Design appropriate electronic circuits, communication modules, software systems and tools to address the defined problem specifications.
- CO3.** Plan and manage the project timeline effectively, including task division, progress tracking, and resource allocation.
- CO4.** Select and utilize suitable hardware platforms, simulation tools, programming environments, and testing equipment for the project.
- CO5.** Document technical progress and present findings clearly through periodic reports and oral presentations, demonstrating individual understanding and contribution.

Evaluation of CA:

- The project will be evaluated at multiple stages periodically and will be graded accordingly by project mentor and or examiner.
- CA evaluations schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by project incharge.